

Introduction

Once a continuous diagnostic test's discrimination is established, a clinical cutpoint must be chosen to turn continuous scores into binary decisions. The optimal cutpoint depends on the trade-off between sensitivity and specificity and on relative costs of false positives vs false negatives.

Prerequisites

Sensitivity and specificity; ROC curves.

Theory

Common criteria: - **Youden's J**: maximise $\text{Sens} + \text{Spec} - 1$. Implicitly assumes equal cost of FN and FP. - **Closest to (0, 1)**: minimise $\sqrt{(1 - \text{Sens})^2 + (1 - \text{Spec})^2}$. - **Cost-weighted**: minimise $c_{FN}(1 - \text{Sens}) \cdot p_D + c_{FP}(1 - \text{Spec}) \cdot (1 - p_D)$ where p_D is prevalence. - **Target specificity (or sensitivity)**: fix one and maximise the other.

Assumptions

Target population has a known prevalence; costs of misclassification are elicited or set by convention; cutpoint will generalise to a new sample.

R Implementation

```
library(pROC); library(cutpointr)

set.seed(2026)
n <- 300
disease <- factor(sample(c(0, 1), n, replace = TRUE, prob = c(0.6, 0.4)))
score <- rnorm(n, mean = ifelse(disease == 1, 1.2, 0))

roc_obj <- roc(response = disease, predictor = score,
               levels = c("0", "1"), direction = "<")
youden_thr <- coords(roc_obj, "best", best.method = "youden",
                    transpose = FALSE)
youden_thr

# cutpointr package: multiple criteria in one call
cp <- cutpointr(data.frame(score, disease),
                x = score, class = disease,
                method = maximize_metric, metric = youden)
summary(cp)
```

Output & Results

Cutpoint at Youden's J and associated sensitivity/specificity. Common cutpoint in this simulation: ~0.6 with sensitivity ~0.7 and specificity ~0.7.

Interpretation

“Maximum Youden's J was achieved at a cutpoint of 0.63 (sensitivity 0.72, specificity 0.71). In a population with 40 % prevalence, this yields PPV 0.62 and NPV 0.80.”

Practical Tips

- Internal cutpoints overfit the training data; cross-validate or use a separate holdout set.
- Clinical cutpoints should be stable, rounded to a meaningful precision, and validated prospectively.
- For screening tests, favour sensitivity; for confirmatory tests, favour specificity.
- Report both the cutpoint and its downstream metrics (Sens, Spec, PPV, NPV, LR+, LR-).
- Decision-curve analysis (Vickers-Elkin) incorporates clinical utility across a range of threshold probabilities.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Diagnostic testing: Se, Sp, PPV, NPV, LR
- Kappa, ICC, Bland–Altman
- Biomarker statistics (Youden, NRI, decision curves)
- TRIPOD-AI, fairness auditing, reproducibility at scale